

Project Report – Stationery Shop CRUD Manager

Name: Rashi Agarwal

Roll Number: 25BCY10248

Course: FLIPPED

1. Title of the Project

Stationery Shop CRUD Manager

2. Introduction

A stationery shop deals with a wide variety of products such as notebooks, pens, files, and other school/office supplies. Managing this inventory manually often leads to errors, outdated stock information, and time-consuming updates.

To solve this, a CRUD-based (Create, Read, Update, Delete) management system is proposed to digitally maintain product details, categorize items, manage stock, and reduce manual workload.

3. Problem Statement

Manual inventory management results in inaccurate stock tracking, duplicate entries, and slower shop operations.

This project aims to build an efficient software system for stationery shop owners to manage products, categories, and stock levels using an easy-to-use CRUD interface.

4. Objectives of the Project

- To develop a clean and user-friendly interface for managing inventory.
- To implement CRUD operations for categories and products.
- To track stock availability in real-time.
- To reduce errors caused by manual entries.
- To maintain updated and accurate information.

5. Scope of the Project

The system covers:

- ✓ Product management
- ✓ Category management
- ✓ Stock handling

✓ Optional billing functionality

✓ CRUD operations for all modules

It does not include online payments, GST invoice generation, or multi-shop management.

6. Functional Requirements

Module 1: Product Management

- Add, view, update, and delete product entries
- Store product ID, name, category, price, and stock quantity

Module 2: Category Management

- Add, view, update, delete categories
- Prevent category deletion if linked to products

Module 3: Stock Management

- Update stock levels
- Low-stock warnings
- Optional stock history

Module 4: Billing Module (Optional)

- Create invoices
- Add products to cart
- Calculate totals
- Auto-deduct stock

7. Non-Functional Requirements

- Usability: Simple UI for non-technical shopkeepers
- Performance: Fast CRUD response
- Security: Limited access for authorized users
- Maintainability: Modular and clean code
- Scalability: New products/categories can be added anytime

- Reliability: Stable and error-free data handling

8. System Architecture

User → UI Layer → Logic Layer → Data Storage (Database/File)

- UI Layer: Data input/output screens
- Logic Layer: CRUD operations
- Data Layer: JSON/CSV/SQL tables storing data

9. UML Diagrams (Text Description)

Use-Case Diagram:

Actor: Admin (Shop Owner)

Use Cases: Manage Products, Manage Categories, Manage Stock, Billing

Class Diagram:

Classes: Product, Category, StockManager, BillingManager (optional), DataHandler

Relationships: Category → Products (One-to-many)

10. Database / File Structure

Categories Table:

- category_id
- category_name

Products Table:

- product_id
- product_name
- category_id
- price
- quantity

Sales Table (Optional):

- sale_id

- total_amount

- date

11. Implementation Details

Technologies Used:

Python / Java / C++

Data stored in JSON / CSV / SQL

Console or GUI-based interface

Key Features Implemented:

- Add, view, edit, delete products

- Manage categories

- Track stock

- Generate bills (optional)

12. Testing Approach

Performed test cases:

- Product addition

- Product deletion

- Category update

- Stock update

- Billing functionality (optional)

Testing methods:

- Manual testing

- Unit testing of core CRUD functions

13. Results & Screenshots

(Insert screenshots later)

14. Challenges Faced

- Maintaining product-category relationship
- Preventing invalid inputs
- Designing intuitive UI
- Keeping data consistent in CRUD operations

15. Learning Outcomes

- Understood CRUD operations in depth
- Practiced modular programming
- Learned system documentation
- Built a complete mini-application
- Improved logical thinking and debugging skills

16. Future Enhancements

- Barcode scanning
- GST invoice printing
- Cloud database integration
- Mobile app version
- Multi-user access with authentication

17. Conclusion

The Stationery Shop CRUD Manager successfully automates inventory management and minimizes errors. It provides a structured way to manage products and stock and can be extended into a complete shop management system in the future.